

Session 2

Introduction to Machine Learning

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What will We Learn Today?

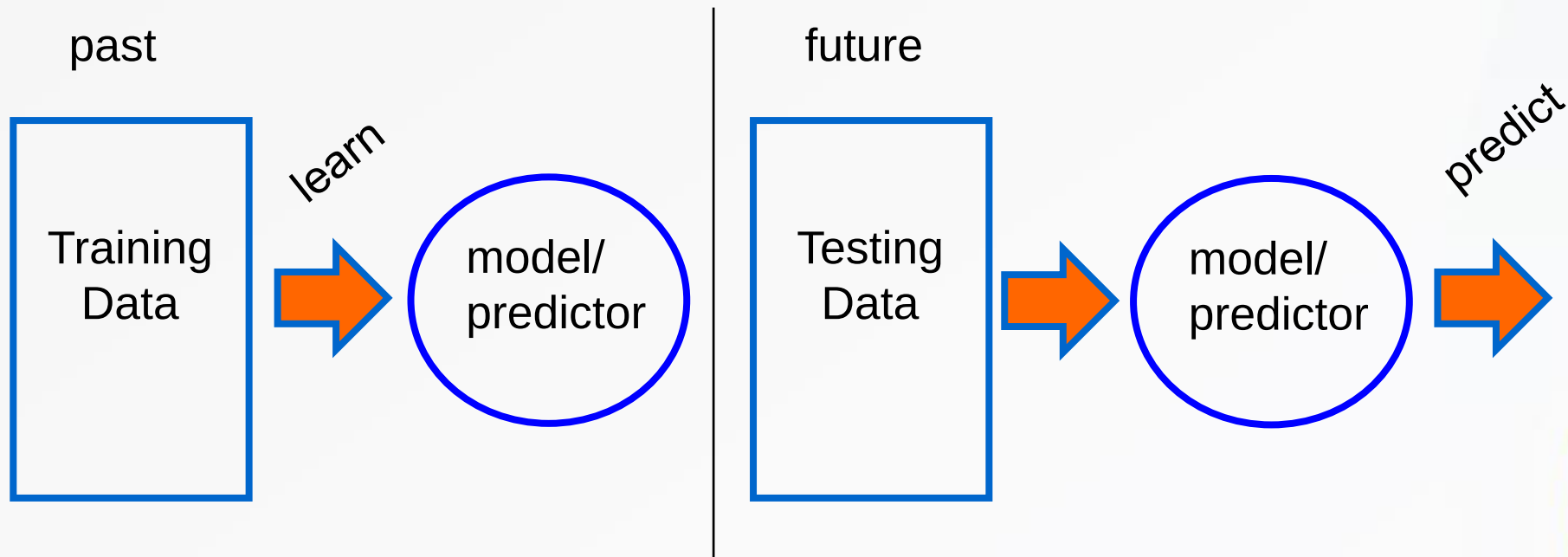
1. Machine Learning
2. ML approaches
3. What is a Good Model?
4. Supervised vs Unsupervised Learning

Machine Learning



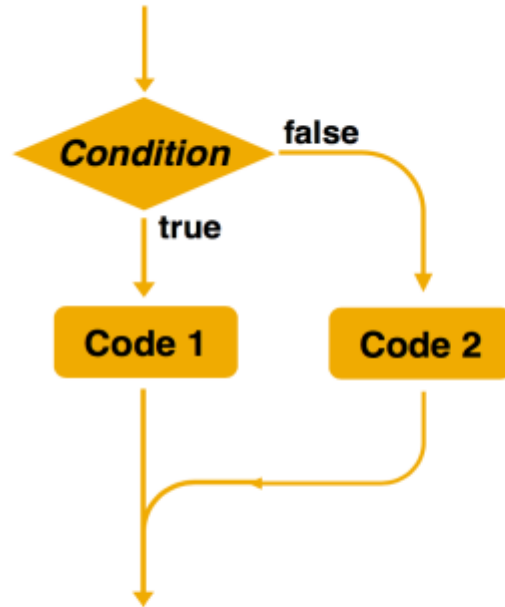
What is Machine Learning

- Cabang kecerdasan buatan (Artificial Intelligence/ AI), yang berkaitan dengan desain dan pengembangan algoritma yang memungkinkan komputer mengembangkan perilaku berdasarkan data empiris.
- Karena kecerdasan membutuhkan pengetahuan, maka komputer perlu memperoleh pengetahuan.



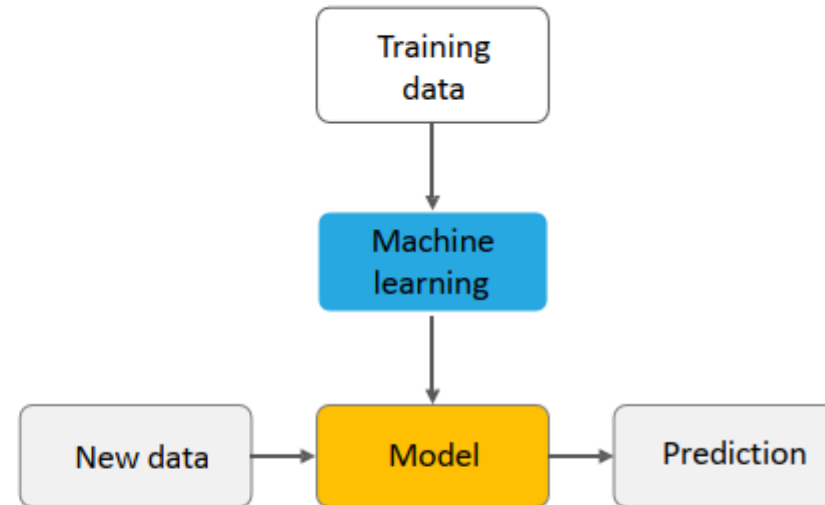
ML vs Traditional Programming

Rule-based algorithms



- Explicit programming is used to solve problems.
- Rules can be manually specified.

Machine learning



- Samples are used for training.
- The decision-making rules are complex or difficult to describe.
- Rules are automatically learned by machines.

ML Approaches

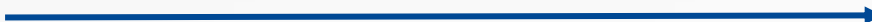


Types of Learning

1. Supervised learning
 - Training data mempunyai target class
 - Classification, regression/ prediction
2. Unsupervised learning
 - Training data tidak mempunyai target class
 - Clustering, association rules
3. Semi-supervised learning
 - Sebagian training data memiliki outputs
4. Reinforcement learning
 - Rewards (hadiah) diberikan ketika agent sukses dalam tugas tertentu,
 - dan dihukum (punishment) ketika salah.



Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

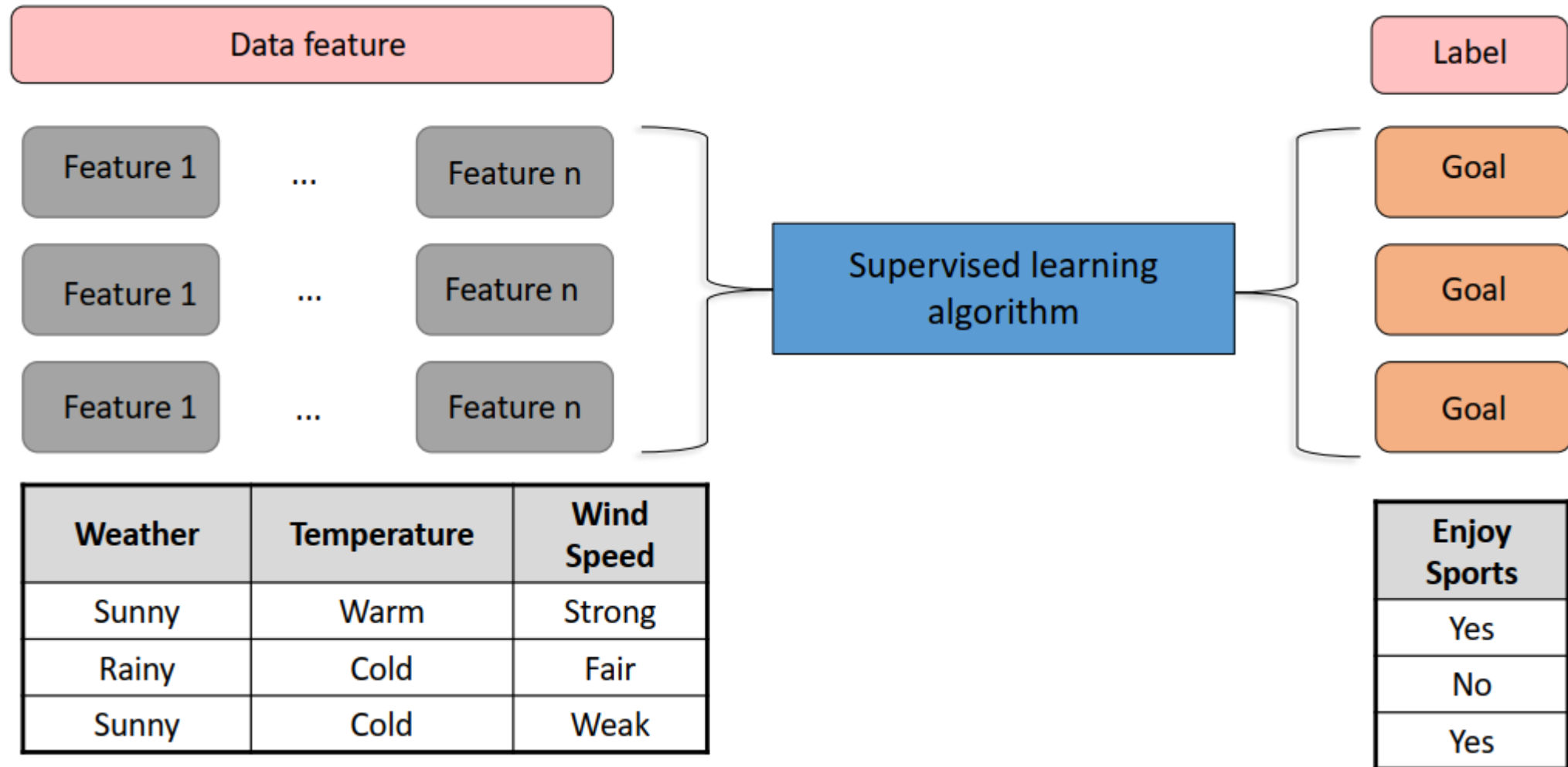


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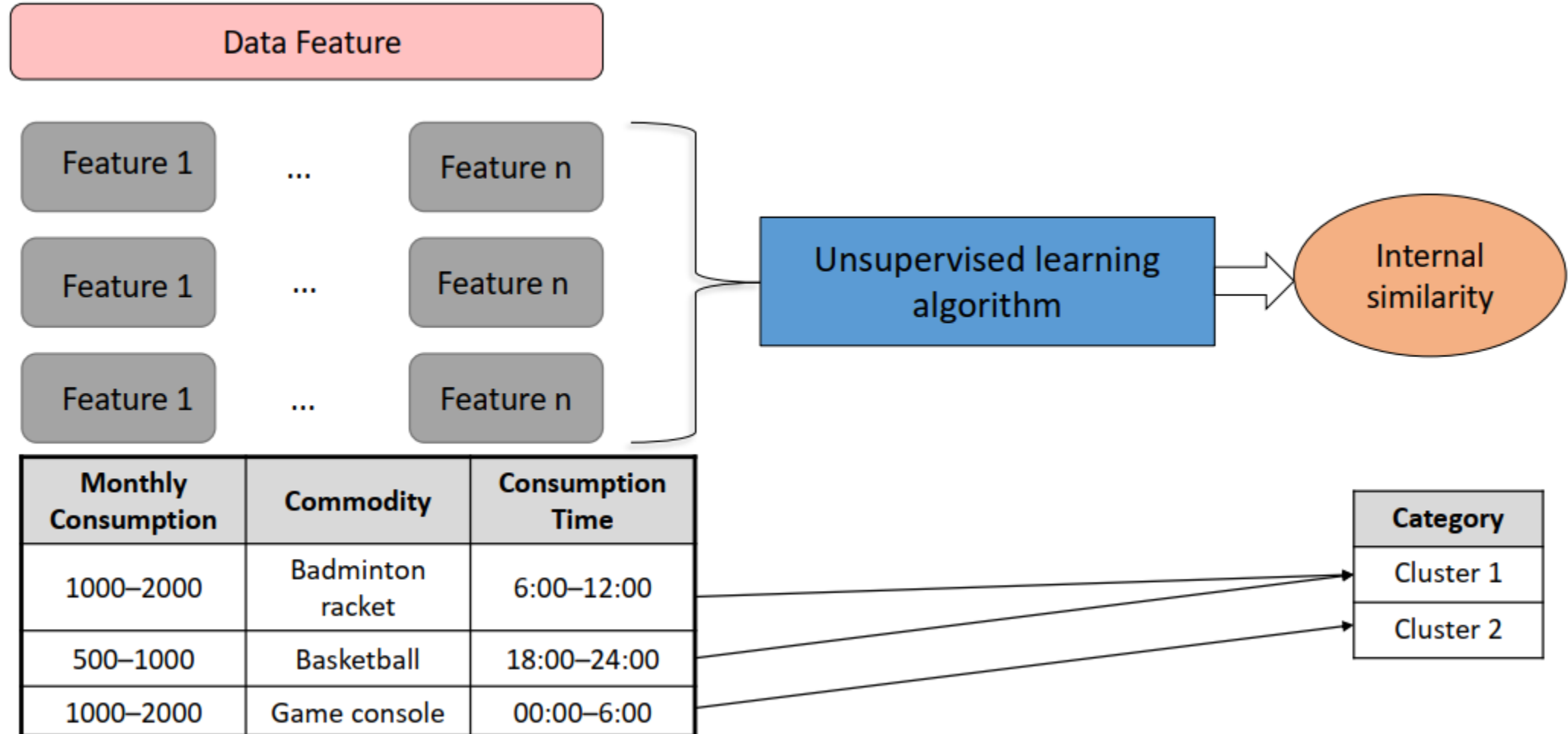


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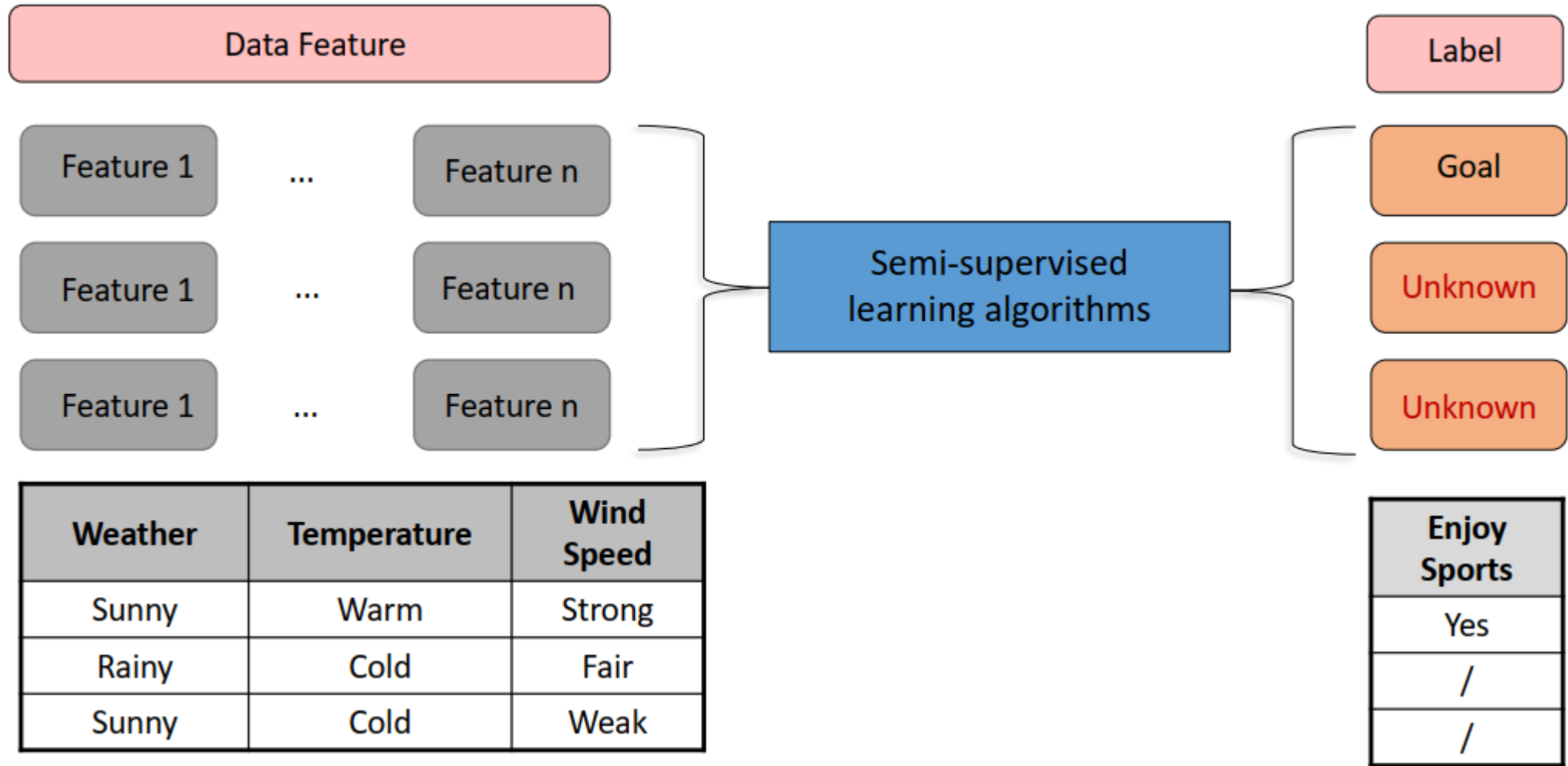
Supervised Learning



Unsupervised Learning

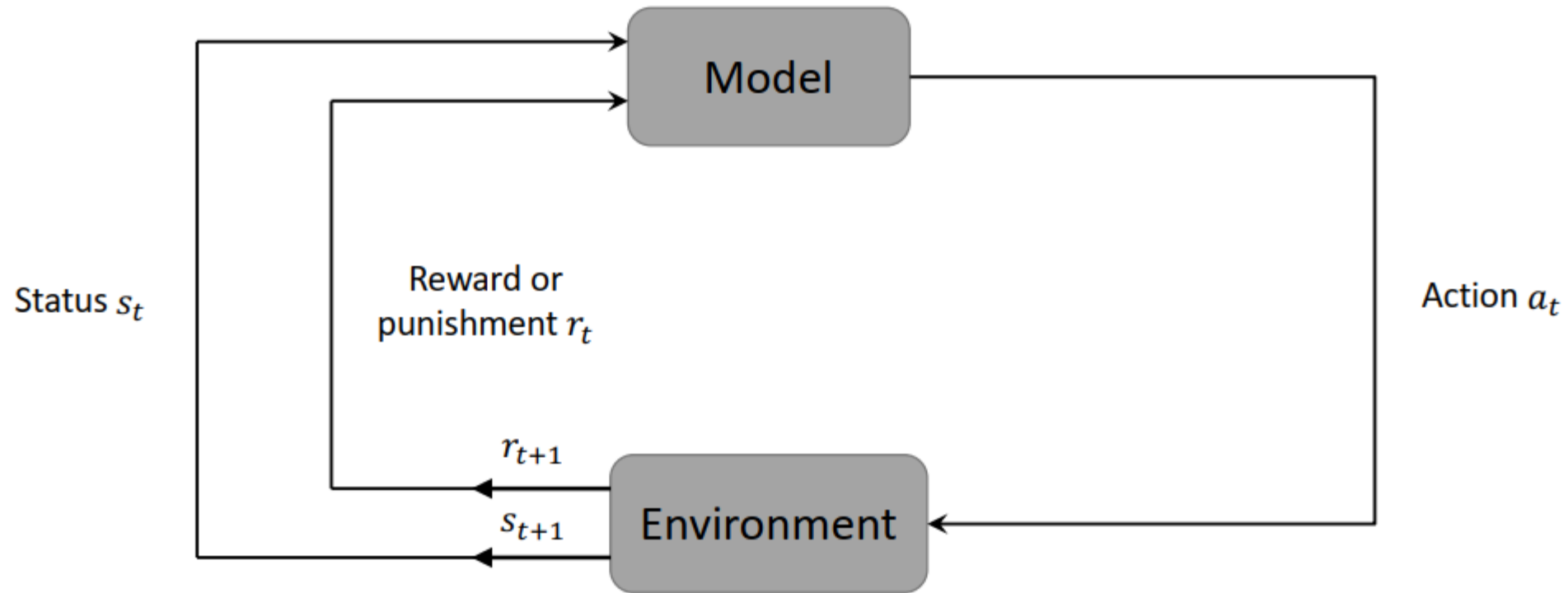


Semi-supervised Learning

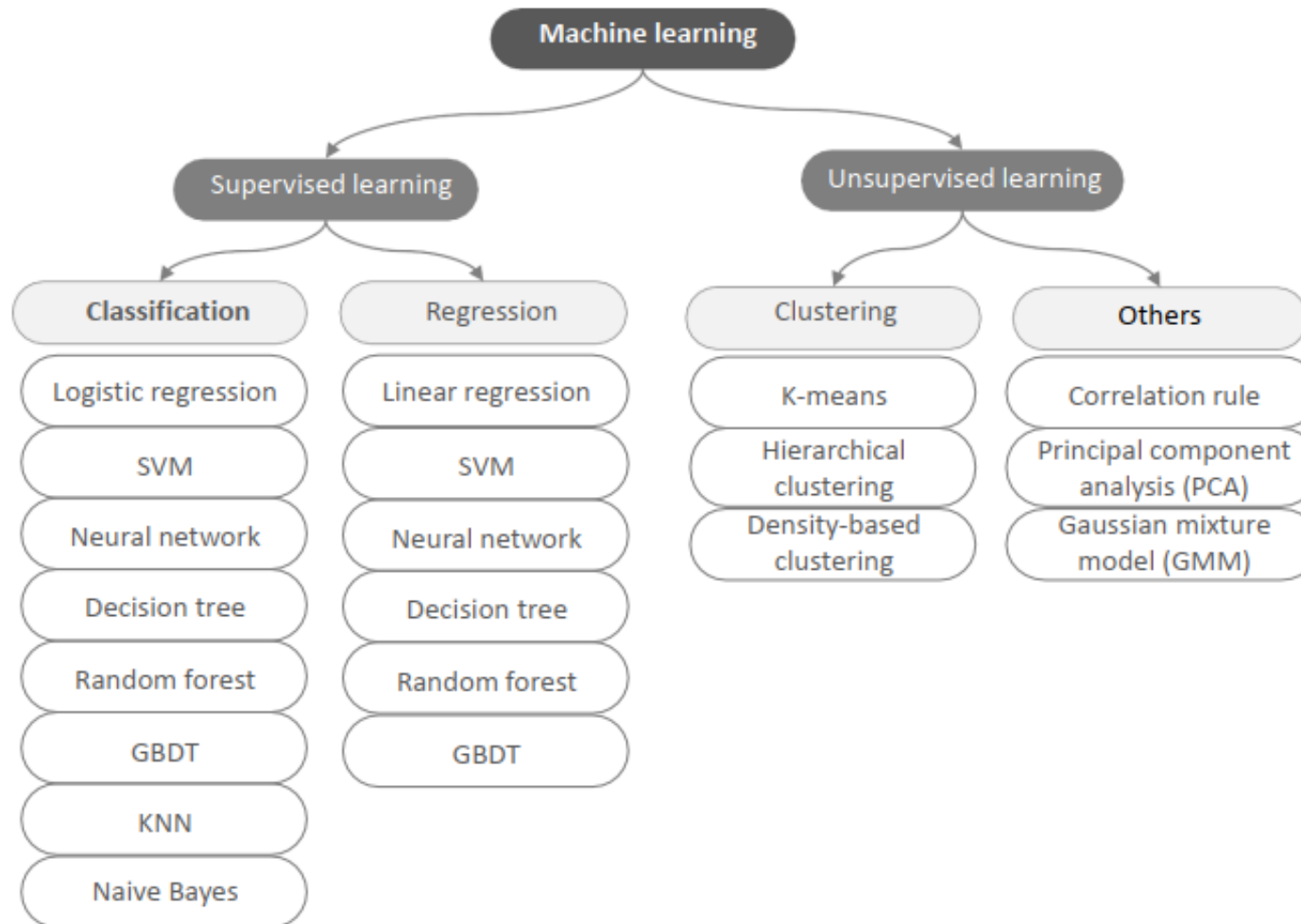


Reinforcement Learning

- The model perceives the environment, takes actions, and makes adjustments and choices based on the status and award or punishment.



Machine Learning Algorithms



Stage in Machine Learning

- Data preprocessing
 - Data cleaning, filling missing value, remove outlier
- Train models
 - Select the algorithm
 - Feature selection and extraction
- Evaluate model
 - Assess performance
 - Model comparison
- Deploy model
 - Apply model to new data
 - Real-time demonstration



Dataset

- Typical dataset form

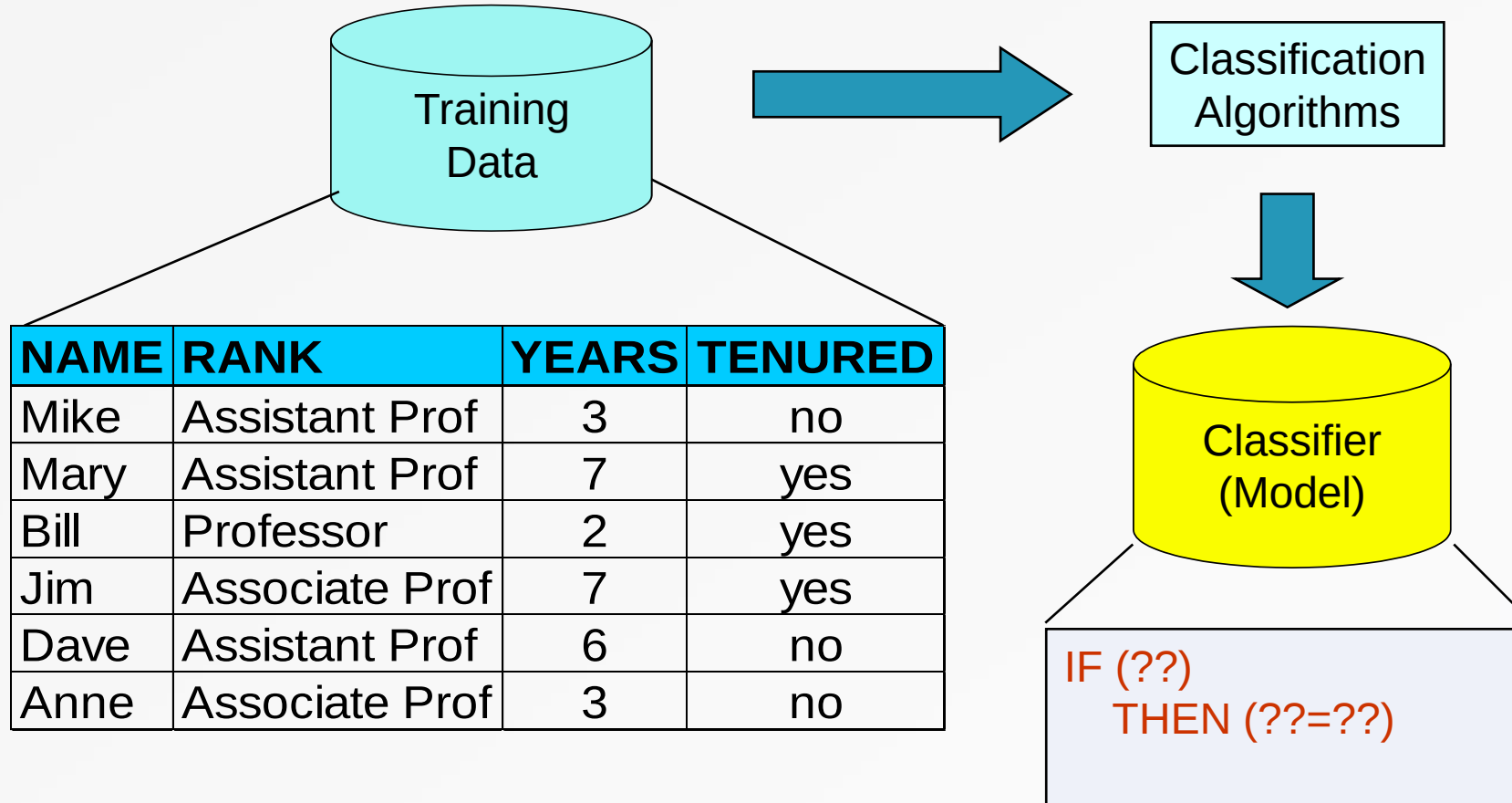
		Feature 1	Feature 2	Feature 3	Label	
		No.	Area	School Districts	Direction	House Price
Training set	1	100	8	South	1000	
	2	120	9	Southwest	1300	
	3	60	6	North	700	
	4	80	9	Southeast	1100	
Test set	5	95	3	South	850	

Why Data Preprocessing?

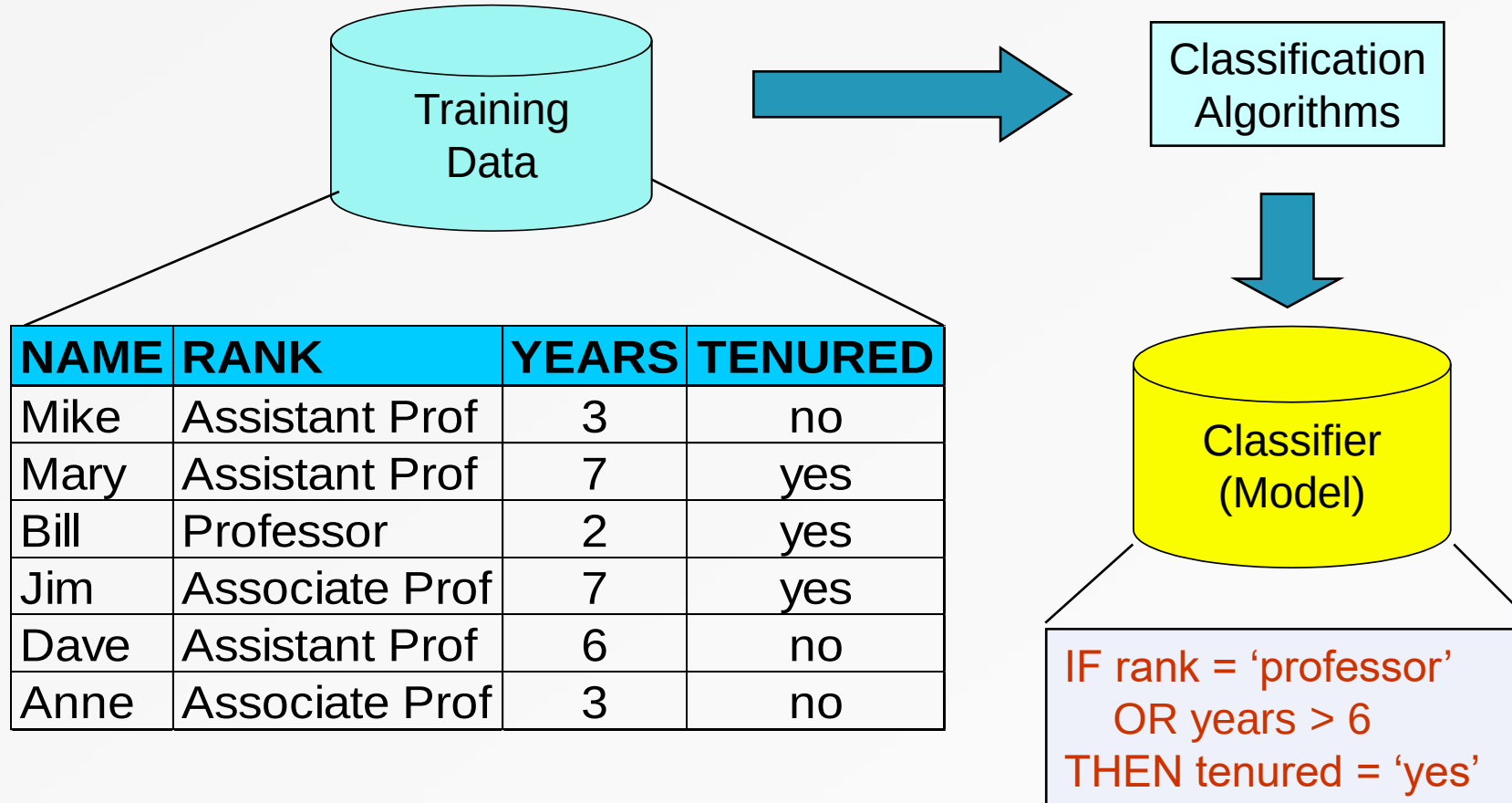
- Data in the real world is dirty
 - **Missing or incomplete**: lacking attribute values,
 - e.g., occupation=" "
 - **Noisy**: containing errors or outliers
 - e.g., Salary="-10"
 - **Inconsistent**: containing discrepancies in codes or names
 - e.g., sex="Girl" vs. sex="Female"
- No quality data, no quality mining results!
 - Quality decisions must be based on quality data

Sex	Age	BMI	DM type	DM duration	FBS	Sys BP	Dias BP	Retinopathy
Male	65	25	II	20	129	130	80	Yes
Male	42	27	II	300	210	140	90	No
Female	31	21	I	11	164	145	80	Yes
Male	70	32	II	29	208	160	100	Yes
Female	54	34	II	6	183	155	95	No
	46	29	II	7	198	160	100	No
Female	16	24	I	-1	250	135	80	No
Male	67	30	II	12	243	165	90	Yes
Female	51	28	II	7	163	130	85	No
Girl	70	36	II	20	250	150	90	Yes
Female	63	35	II	14	203	160	110	No
Male	44	39	II	3	149	140	90	No
Boy	51	24	II	9	160	155	80	No
Male	27	19	I	5	170	140	90	No

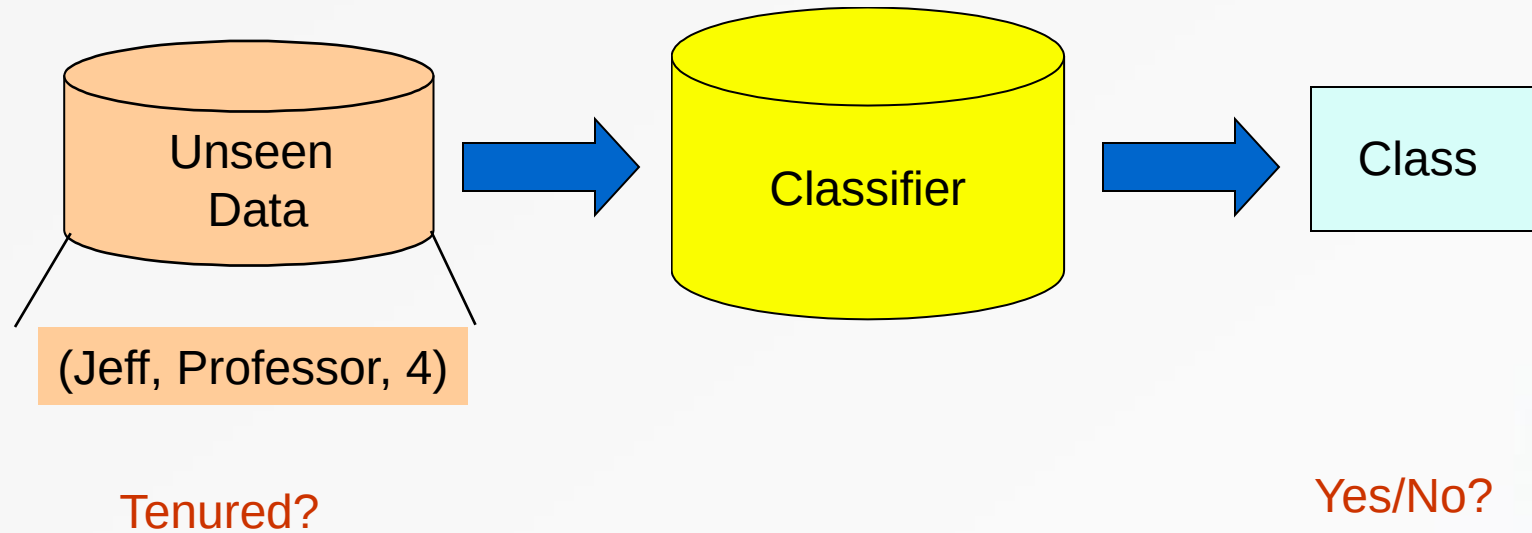
Supervised Learning - Learning Phase



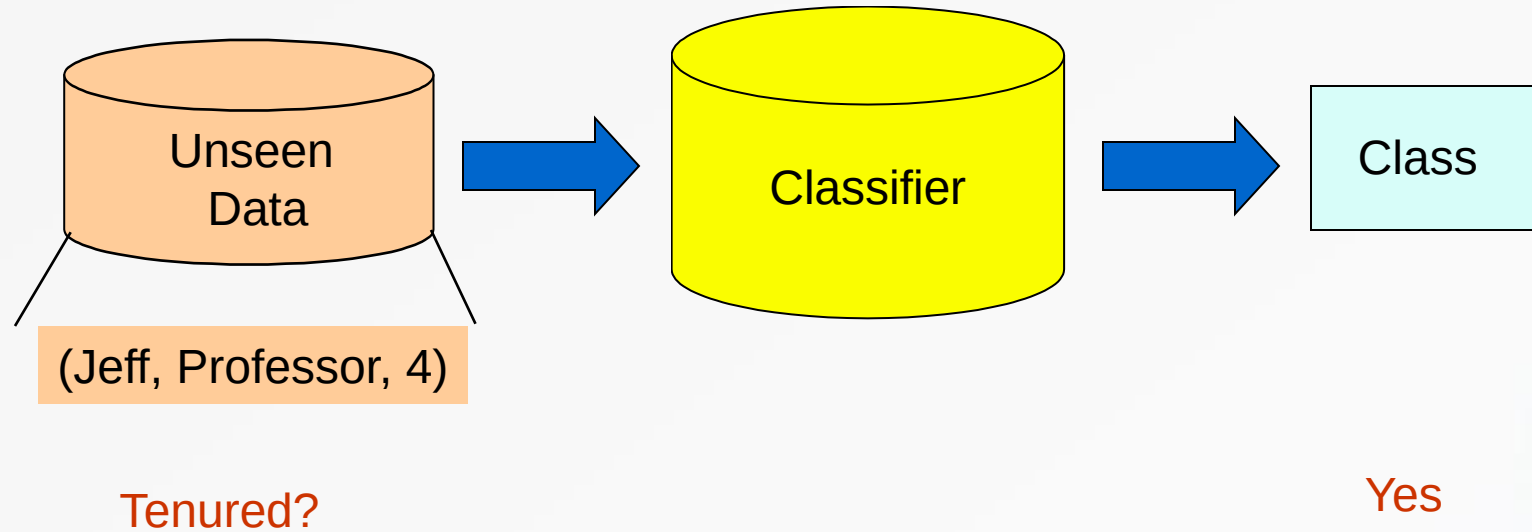
Supervised Learning - Learning Phase



Supervised Learning - Prediction Phase



Supervised Learning - Prediction Phase



What is a Good Model?



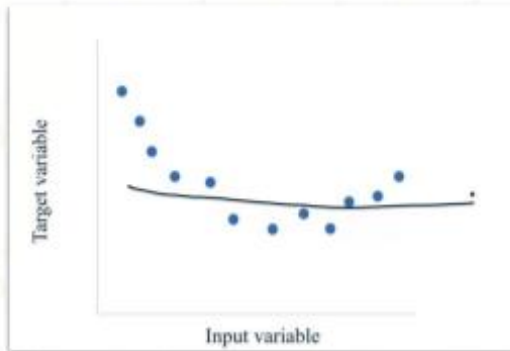
- **Generalization capability**
Can it accurately predict the actual service data?
- **Interpretability**
Is the prediction result easy to interpret?
- **Prediction speed**
How long does it take to predict each piece of data?
- **Practicability**
Is the prediction rate still acceptable when the service volume increases with a huge data volume?

Model Validity

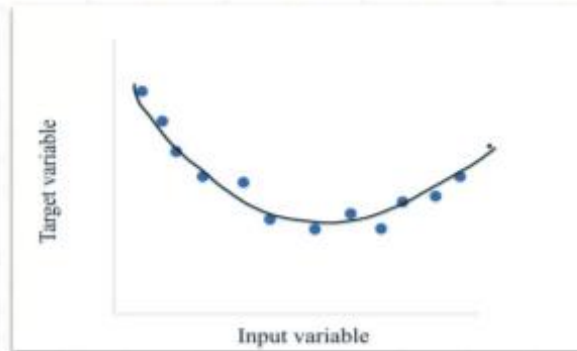
- The goal of machine learning is that the model obtained after learning should perform well on new samples, not just on samples used for training.
- The capability of applying a model to new samples is called generalization or robustness

Underfitting and overfitting

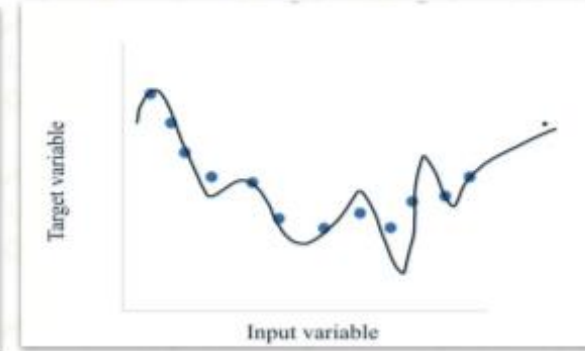
Underfitting in Regression



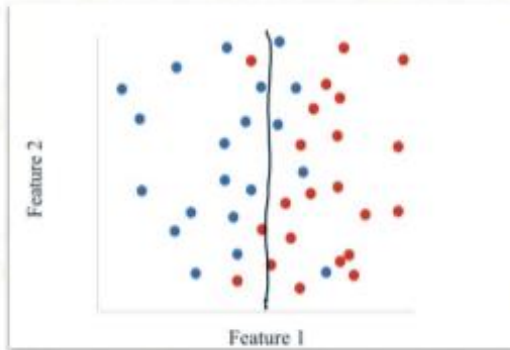
Good Model in Regression



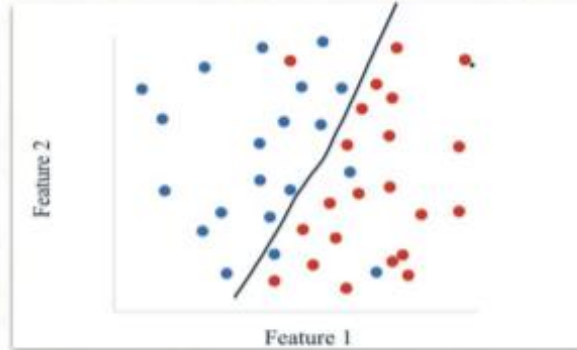
Overfitting in Regression



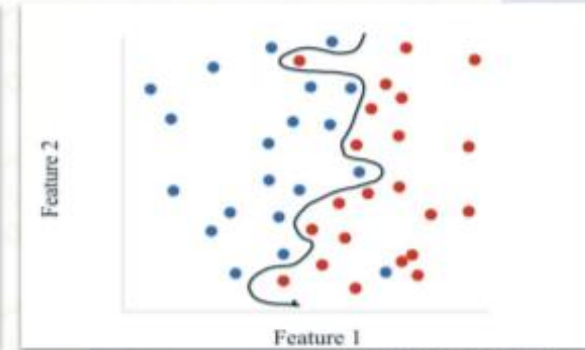
Underfitting in Classification



Good Model in Classification

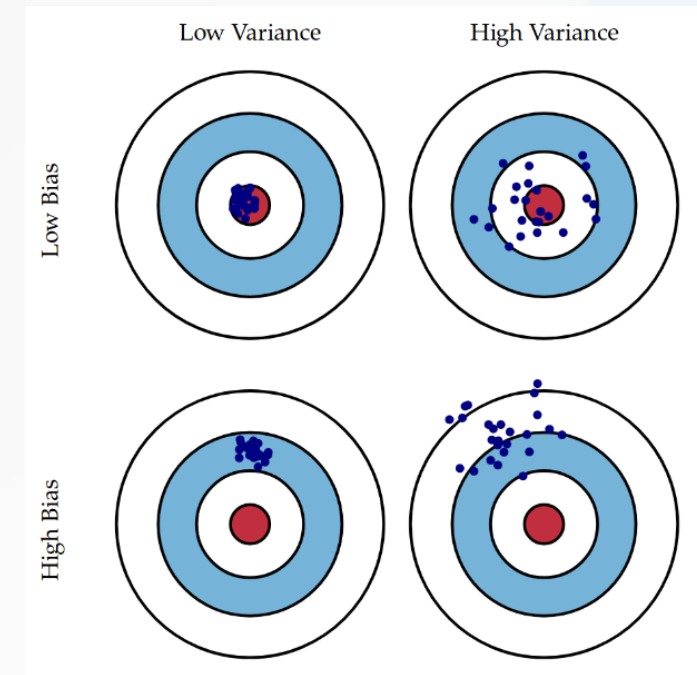


Overfitting in Classification

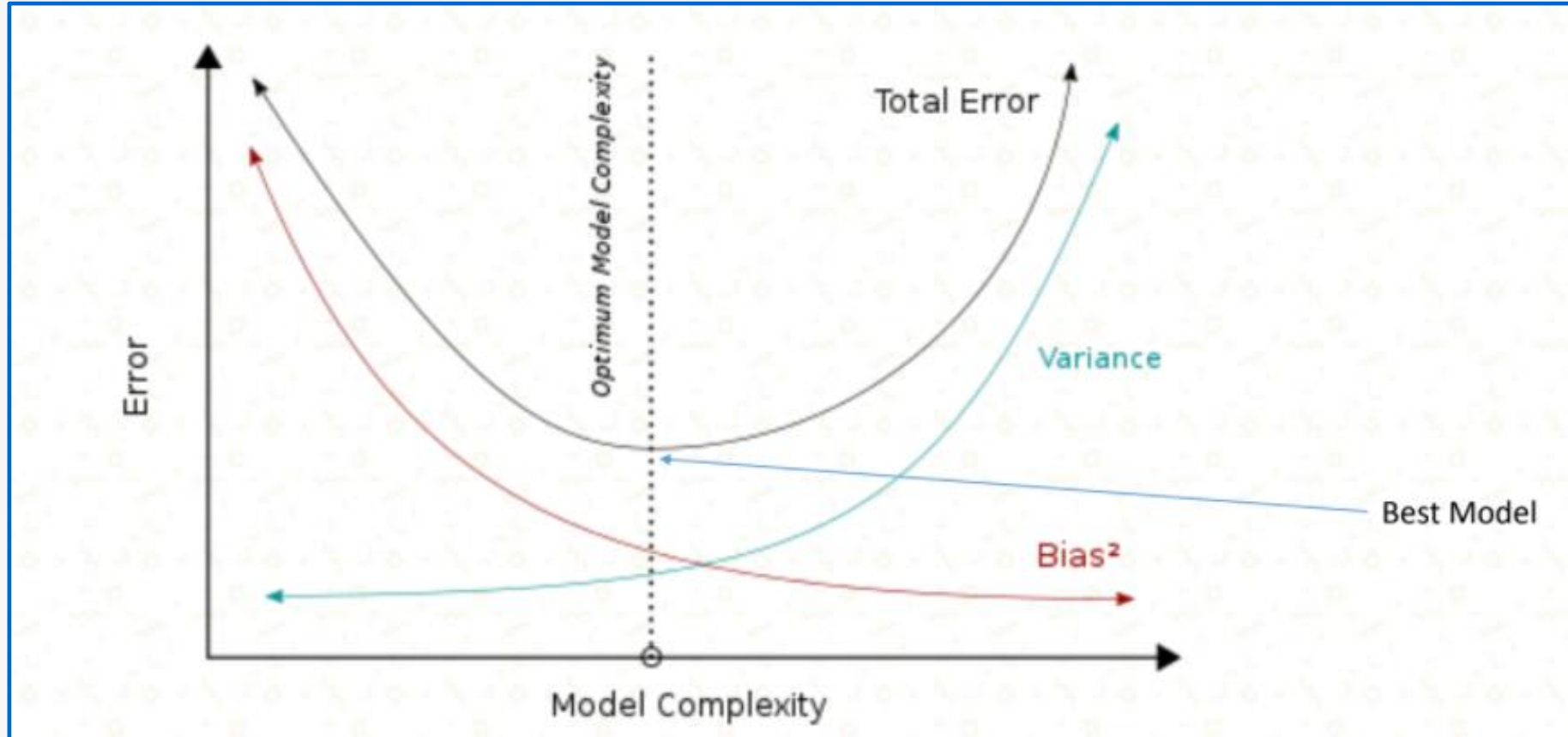


Bias and variance

- **Bias**
- Bias adalah perbedaan antara rata rata hasil prediksi dari model ML yang kita develop dengan data nilai yang sebenarnya.
- Bias yang tinggi dikarenakan dalam pembangunan model ML, dilakukan terlalu sederhana (oversimplified).
- **Variance**
- Variance adalah variabel dari prediksi yang memberikan kita informasi perserbaran data hasil prediksi.
- Model yang memiliki variance tinggi memiliki korelasi kuat hanya pada training set, sehingga akan berkinerja baik pada training data saja.



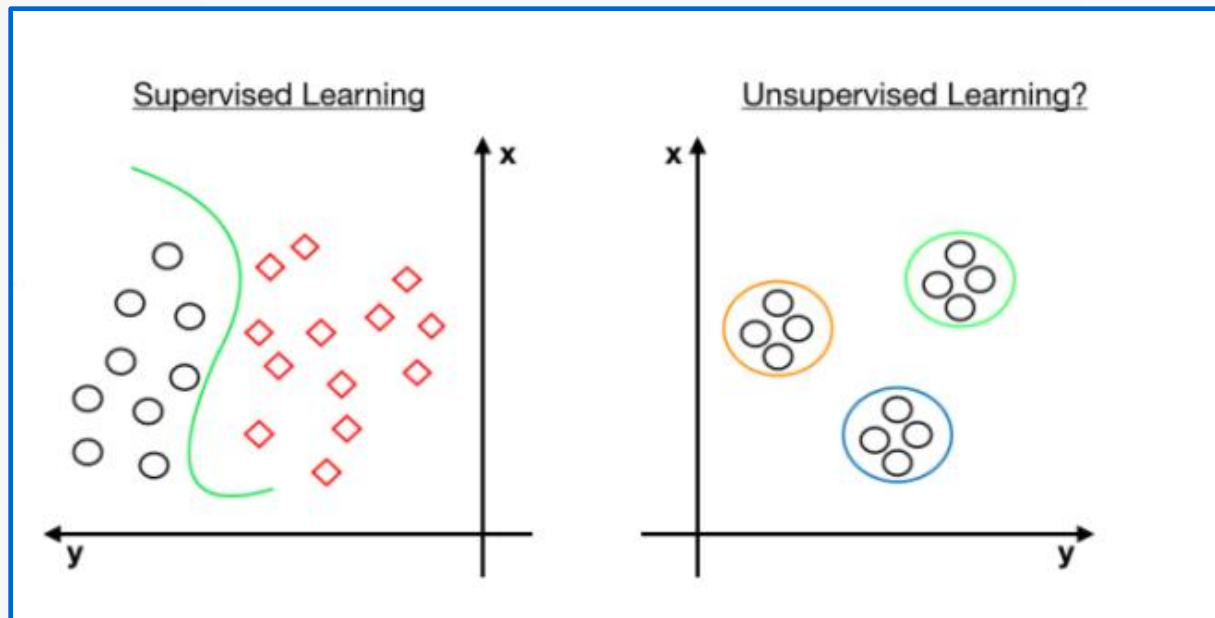
Bias variance tradeoff



Supervised vs Unsupervised Learning

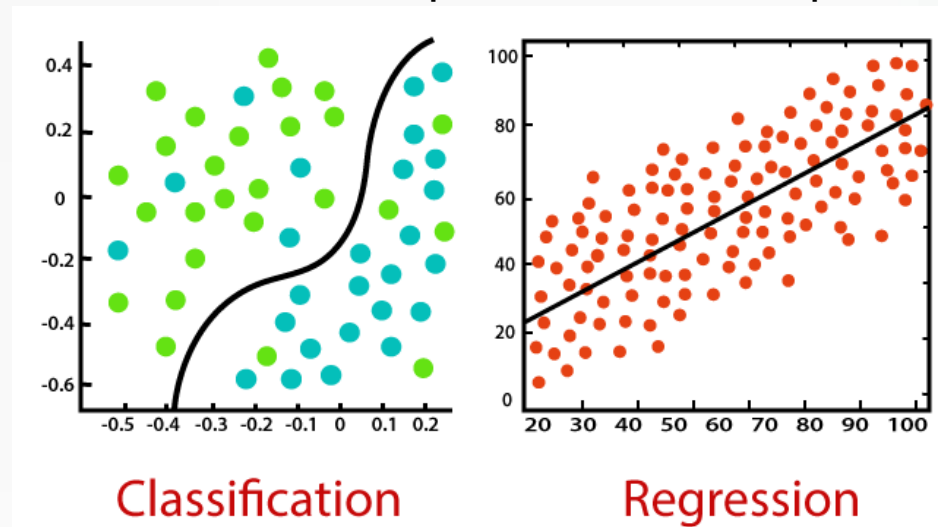
Supervised vs Unsupervised Learning

- Supervised = Mempelajari data untuk memprediksi output.
 - Kita tahu target label, sehingga kita membuat model untuk memprediksi label.
- Unsupervised = Menemukan pattern/ characteristic dari data.
 - Kita tidak mengetahui target label, sehingga kita membuat model yang mencoba mengelompokkan data.



Supervised Learning

- *Classification* (klasifikasi) = metode yang menarik beberapa kesimpulan dari nilai input yang diberikan pada saat training dan kemudian akan memprediksi label/kelas untuk data baru.
- *Regression* (regresi) = metode yang mencoba untuk menentukan kekuatan dan karakter hubungan antara satu variabel dependen dan serangkaian variabel lainnya (variabel independen).
- Algoritma regresi = nilai kontinu (seperti harga, gaji, usia, dll).
- Algoritma klasifikasi = nilai diskrit (seperti stroke atau normal, spam atau bukan spam, dll)
- *Both are supervised learning*



Classification, regression, clustering

price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	condition	grade	sqft_above	sqft_basement	yr_built
221900.0	3	1.00	1180	5650	1.0	0	0	3	7	1180	0	1955
538000.0	3	2.25	2570	7242	2.0	0	0	3	7	2170	400	1951
180000.0	2	1.00	770	10000	1.0	0	0	3	6	770	0	1933
604000.0	4	3.00	1960	5000	1.0	0	0	5	7	1050	910	1965
510000.0	3	2.00	1680	8080	1.0	0	0	3	8	1680	0	1987

Regression (house price dataset)

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke
0	9046	Male	67.0	0	1	Yes	Private	Urban	228.69	36.6	formerly smoked	1
1	51676	Female	61.0	0	0	Yes	Self-employed	Rural	202.21	NaN	never smoked	1
2	31112	Male	80.0	0	1	Yes	Private	Rural	105.92	32.5	never smoked	1
3	60182	Female	49.0	0	0	Yes	Private	Urban	171.23	34.4	smokes	1
4	1665	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.0	never smoked	1
...	...	...	...	...	...	...	...	...	...	...	...	...
5105	18234	Female	80.0	1	0	Yes	Private	Urban	83.75	NaN	never smoked	0
5106	44873	Female	81.0	0	0	Yes	Self-employed	Urban	125.20	40.0	never smoked	0
5107	19723	Female	35.0	0	0	Yes	Self-employed	Rural	82.99	30.6	never smoked	0
5108	37544	Male	51.0	0	0	Yes	Private	Rural	166.29	25.6	formerly smoked	0
5109	44679	Female	44.0	0	0	Yes	Govt_job	Urban	85.28	26.2	Unknown	0

Classification (stroke dataset)

	ID	Sex	Marital status	Age	Education	Income	Occupation	stroke
0	100000001	0	0	67	2	124670	1	1
1	100000002	1	1	22	1	150773	1	1
2	100000003	0	0	49	1	89210	0	0
3	100000004	0	0	45	1	171565	1	1
4	100000005	0	0	53	1	149031	1	1

Clustering (customer dataset)

Let's practice

Thank You